



AMC ENGINEERING COLLEGE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Capacity Planning for IT

Sub Code:BCS456B

MODULE-3

1.Which of the following are types of requirements and capacity goals interrelated with performance, availability, and reliability?

- A. External service monitoring
- B. Business requirements
- C. User expectations
- D. All of the above

Answer: D. All of the above

2.Which of the following metrics are used by commercial services to monitor site performance?

- A. Page load time
- B. Time to first byte (TTFB)
- C. Secure Sockets Layer (SSL) time
- D. All of the above

Answer: D. All of the above

3.Why is it important to understand what monitoring services measure and how to interpret the numbers they generate?

- A. To ensure the statistics can be used to enforce SLAs
- B. To be aware of how web pages are being requested

- C. To determine if they are simulating human users
- D. All of the above

Answer: D. All of the above

4. What should be considered when using a monitoring service?
- A. Whether they simulate human users
 - B. If they cache objects like a normal web browser
 - C. If time spent is due to network transfer versus server time
 - D. All of the above

Answer: D. All of the above

5. What type of intent affects user interaction with a website or mobile app in the context of search?
- A. Navigational
 - B. Informational
 - C. Commercial
 - D. All of the above

Answer: D. All of the above

6. Which of the following monitoring services are considered enterprise-level?

- A. PingDom
- B. SiteUptime
- C. Keynote
- D. Alertra

Answer: C. Keynote

7.What can potentially induce a performance penalty or affect availability in a distributed database?

- A. Caching objects
- B. Guaranteeing consistency
- C. User expectations
- D. Network transfer time

Answer: B. Guaranteeing consistency

8.Who might become addicted to qualitative assessments of operations based on metrics used for capacity planning or site performance measurement?

- A. CFOs
- B. CTOs
- C. CEOs
- D. All of the above

Answer: D. All of the above

9.What is an SLA primarily used for in a business context?

- A. To define software requirements
- B. To make business people comfortable and establish performance metrics**
- C. To outline project deadlines
- D. To describe company policies

Answer-B

10.What is a primary reason businesses tie revenue streams to unfettered access to an API?

- A. To ensure security
- B. To guarantee availability and performance of the API
- C. To facilitate user registration
- D. To manage marketing efforts

Answer: B. To guarantee availability and performance of the API

11. Which of the following metrics are proposed to quantify perceived performance of a website or mobile app?

- A. Speed index
- B. Render start
- C. DOM interactive
- D. All of the above

Answer: D. All of the above

12. Which factor is NOT directly related to capacity but can still affect the perceived performance of a website?

- A. High-quality images or videos
- B. Number of servers
- C. Page weight
- D. Construction of the web page

Answer: B. Number of servers

13. Why is determining the root cause of perceived slowness of a web page easier in monolithic architectures than in Service-Oriented Architecture (SOA) or microservice architecture (MSA)?

- a. Monolithic architectures are more modern
- b. Monolithic architectures use more third-party vendors
- c. SOA and MSA do not use high-quality images
- d. SOA and MSA have complex interactions between different microservices

Answer: d). SOA and MSA have complex interactions between different microservices

14. What might a shipping company contract permit in terms of API call rate, according to the example provided?

- A. 1 API call per minute

- B. 10 API calls per second
- C. 5 API calls per minute
- D. 20 API calls per second

Answer: B. 10 API calls per second

15. Which of the following is NOT one of the User Timing “Standard Mark Names” mentioned?

- A. mark_fully_loaded
- B. mark_time_to_user_action
- C. mark_page_weight
- D. mark_above_the_fold

Answer: C. mark_page_weight

16. When might a website reach intolerable amounts of latency?

- A. At low traffic volumes
- B. When the server CPU is underutilized
- C. At high traffic volumes before system-level metrics raise a flag
- D. When no third-party services are used

Answer: C. At high traffic volumes before system-level metrics raise a flag

17. What could be a cause of perceived slowness of a web page besides lack of capacity?

- A. Heavy page weight
- B. High disk I/O
- C. Insufficient number of servers
- D. Poor DNS management

Answer: B. To make business people comfortable and establish performance metrics

17. What does an SLA typically cover for websites?

- A. Security and user interface design
- B. Availability and performance
- C. Marketing and sales
- D. Development and testing

Answer: B. Availability and performance

18. What does a 99.9% uptime SLA mean in terms of allowable downtime in a month?

- A. 10 minutes
- B. 30 minutes
- C. 43.2 minutes
- D. 60 minutes

Answer: C. 43.2 minutes

19. According to Table 2-1, what is the acceptable downtime per year for an SLA of 99.99%?

- A. 8 hours, 45 minutes, 36 seconds
- B. 52 minutes, 33 seconds
- C. 5 minutes, 15 seconds
- D. 32 seconds

Answer: B. 52 minutes, 33 seconds

20. What term refers to an availability of 99.999%?

- A. Three-nines
- B. Four-nines
- C. Five-nines
- D. Six-nines

Answer: C. Five-nines

21.If a site generates \$3,000 worth of sales every minute, how much revenue could potentially be lost during a 10-minute downtime, assuming no customer recovery?

- A. \$30,000
- B. \$15,000
- C. \$45,000
- D. \$3,000

Answer: A. \$30,000

22.What might be a more realistic impact of a brief service outage on revenue?

- A. Immediate loss of sales with no recovery
- B. Increased revenue after the outage
- C. Customers picking up where they left off and minimal impact on earnings
- D. Permanent loss of all potential sales during the outage

Answer: C. Customers picking up where they left off and minimal impact on earnings

23.What is a common usage of the term "five-nines" in SLA discussions?

- A. To describe a feature set
- B. To indicate high availability of a site or system
- C. To refer to the number of servers
- D. To measure bandwidth capacity

Answer: B. To indicate high availability of a site or system

24.What is the primary purpose of architecture in backend systems?

- a. To design the layout of backend pieces for capacity planning
- b. To enhance security

c. To reduce the cost of hardware

d. To simplify the user interface

Answer: a. To design the layout of backend pieces for capacity planning

25. Why should architecture be designed to easily split into parts that perform discrete tasks? A. To reduce power consumption

B. To facilitate measurement and rapid response to changing conditions

C. To improve the aesthetic design of the system

D. To increase the number of servers

Answer: B. To facilitate measurement and rapid response to changing conditions

26. What is a key benefit of using Microservices Architecture (MSA)? A. It increases the complexity of the system

B. It allows for quicker changes in response to bugs or feature requests

C. It reduces the need for monitoring

D. It eliminates the need for containerization

Answer: B. It allows for quicker changes in response to bugs or feature requests

27. In a database-driven web example, what is the main difficulty with having the web server and database on the same hardware server? A. Increased security risks

B. Difficulty in distinguishing system statistics for different components

C. Higher cost of maintenance

D. Lower overall performance

Answer: B. Difficulty in distinguishing system statistics for different components

28. What should be measured to establish when to add more capacity to a split architecture?

A. The number of user logins per hour

B. The size of the web pages being served

C. The impact of increased queries-per-second on disk utilization, I/O wait, RAM usage, and CPU usage

D. The frequency of software updates

Answer: C. The impact of increased queries-per-second on disk utilization, I/O wait, RAM usage, and CPU usage

29. Which of the following is a metric that can be used to measure web server performance?

- A. User satisfaction rate
- B. Number of software bugs
- C. Requests-per-second
- D. Number of user reviews

Answer: C. Requests-per-second

30. What is the 'ceiling' in the context of architecture and capacity planning?

- A. The maximum number of users a system can handle
- B. The critical level of a resource that cannot be crossed without failure or SLA violation
- C. The highest level of security implemented in a system
- D. The total cost of running the system

Answer: B. The critical level of a resource that cannot be crossed without failure or SLA violation

31. Why do public clouds offer a wide range of instance types?

- A. To cater to different budgets of users
- B. To provide a variety of operating systems
- C. Because different services exert different resource pressure
- D. To offer various levels of security

Answer: C. Because different services exert different resource pressure

32. What is the purpose of determining the ceiling for each piece of architecture?

- A. To find out the maximum revenue potential
- B. To understand when to upgrade hardware
- C. To prevent failures or SLA violations by not exceeding resource limits
- D. To calculate the total system cost

Answer: C. To prevent failures or SLA violations by not exceeding resource limits

33. What does the "small pieces, loosely joined" theory suggest in architecture design?

- A. Combining all services into a single large service
- B. Dividing labor into smaller, independent units that are loosely connected
- C. Reducing the number of servers
- D. Simplifying the user interface

Answer: B. Dividing labor into smaller, independent units that are loosely connected

34. What can simple architectural changes help you understand? A. The total cost of ownership

- B. The security vulnerabilities in the system
- C. How the capacity is being used for different purposes
- D. The user satisfaction rate

Answer: C. How the capacity is being used for different purposes

35. When armed with current ceilings, what can you begin to assemble? A. A new user interface

- B. A security plan
- C. The capacity plan
- D. A marketing strategy

Answer: C. The capacity plan

:

36. What is vertical scaling?

- A. Adding more servers to share the load
- B. Increasing the resources within a single server
- C. Using cloud services for additional capacity
- D. Reducing the number of servers in use

Answer: B. Increasing the resources within a single server

37. What is horizontal scaling?

- A. Adding more resources to a single server
- B. Decreasing the server load by reducing user traffic
- C. Adding more servers to share the load

D. Using more powerful servers to handle increased load

Answer: C. Adding more servers to share the load

38. What is diagonal scaling?

- A. Vertically scaling horizontally scaled nodes
- B. Scaling only the CPU of existing servers
- C. Scaling only the memory of existing servers
- D. Combining both vertical and horizontal scaling on new servers

Answer: A. Vertically scaling horizontally scaled nodes

39. What is a potential disadvantage of relying solely on vertical scaling?

- A. Increased complexity in managing multiple servers
- B. Higher costs and the risk of a single point of failure
- C. Difficulty in measuring individual server performance
- D. Lower CPU and memory utilization

Answer: B. Higher costs and the risk of a single point of failure

40. In the Yahoo! example, how many dual-quad core machines were used to replace the 67 dual-CPU machines?

- A. 20
- B. 18
- C. 15
- D. 25

Answer: B. 18

41. What was the primary benefit of diagonal scaling in the Yahoo! example?

- A. Decreased disk usage

- B. Increased network bandwidth
- C. Reduced load average and power consumption
- D. Improved security

Answer: C. Reduced load average and power consumption

42. What does the term "SPOF" stand for?

- A. Single Point of Failure
- B. Scalable Point of Function
- C. Standard Protocol of Functionality
- D. System Performance Optimization Framework

Answer: A. Single Point of Failure

43. Why is horizontal scaling considered advantageous for web applications?

- A. It increases the cost efficiency of vertical scaling
- B. It allows for easy synchronization between nodes
- C. It handles various levels of database abstraction and distribution
- D. It simplifies security management

Answer: C. It handles various levels of database abstraction and distribution

44. What does Disaster Recovery (DR) primarily focus on?

- A. Enhancing website security
- B. Saving business operations after a catastrophe
- C. Improving website performance
- D. Reducing server costs

Answer: B. Saving business operations after a catastrophe

45. Which of the following is NOT typically considered a catastrophe for DR purposes?

- A. Datacenter power outages

- B. Earthquakes
- C. Single server failure
- D. Cooling outages

Answer: C. Single server failure

46. Disaster Recovery (DR) is a part of which larger logistical plan?

- A. Vertical Scaling
- B. Business Continuity Planning (BCP)
- C. Capacity Management
- D. Horizontal Scaling

Answer: B. Business Continuity Planning (BCP)

47. What is a common solution for ensuring Disaster Recovery?

- A. Increasing the resources within a single server
- B. Deploying complete architectures in multiple physical locations
- C. Using only cloud services
- D. Reducing the number of servers

Answer: B. Deploying complete architectures in multiple physical locations

48. What is a significant concern when implementing DR plans?

- A. Simplifying code deployment
- B. Multiplying infrastructure costs
- C. Enhancing server performance
- D. Reducing network bandwidth

Answer: B. Multiplying infrastructure costs

49. Why is recognizing trends important for capacity planning?

- A. To reduce server costs
- B. To identify the fastest servers
- C. To understand resource needs and recurring patterns over time
- D. To minimize network traffic

Answer: C. To understand resource needs and recurring patterns over time

50.What can historical data on capacity help you analyze?

- A. Hardware compatibility
- B. Resource trends and recurring patterns
- C. Server locations
- D. Network speed

Answer: B. Resource trends and recurring patterns

51.What is the fundamental principle of capacity planning mentioned in the text?

- A. Estimating server costs
- B. Using ceiling values and historical data for predictions
- C. Reducing hardware redundancy
- D. Increasing network speed

Answer: B. Using ceiling values and historical data for predictions

52.In the context of the provided text, what does adding a trendline in Excel help with?

- A. Reducing server load
- B. Forecasting future data based on historical trends
- C. Minimizing network traffic
- D. Increasing storage capacity

Answer: B. Forecasting future data based on historical trends

53.What does the process of curve fitting involve?

- A. Finding a mathematical equation that mimics the data
- B. Reducing the number of servers
- C. Enhancing network speed
- D. Increasing CPU performance

Answer: A. Finding a mathematical equation that mimics the data

54.What metric was used to predict disk I/O wait times for databases at Flickr?

- A. Number of users
- B. Number of photos
- C. Ratio of photos-to-users
- D. Server uptime

Answer: C. Ratio of photos-to-users

55.What architectural change at Flickr helped manage the growing data load?

- A. Implementing a monolithic database
- B. Increasing server CPU capacity
- C. Sharding the database
- D. Reducing network bandwidth

Answer: C. Sharding the database

56.What is the danger of making forecasts with small datasets?

- A. Lack of accuracy and potential seasonal variations
- B. Increased network traffic
- C. Reduced server performance
- D. Higher hardware costs

Answer: A. Lack of accuracy and potential seasonal variations

57. What is one advantage of using Excel macros for automating the forecasting process?

- A. Reducing server maintenance
- B. Integrating forecasts into dashboards and databases
- C. Increasing CPU performance
- D. Minimizing network traffic

Answer: B. Integrating forecasts into dashboards and databases

58. What is the primary purpose of the open-source program fityk?

- A. Enhancing server speed
- B. Curve-fitting equations to arbitrary data
- C. Reducing data storage
- D. Increasing network bandwidth

Answer: B. Curve-fitting equations to arbitrary data

59. How can fityk be used to automate curve fitting and forecasting?

- A. By increasing server capacity
- B. By using the command-line version called cfityk
- C. By reducing network latency
- D. By increasing disk space

Answer: B. By using the command-line version called cfityk

60. What does the command file used by cfityk consist of?

- A. Server configurations
- B. A script of actions choreographed in the GUI

C. Network traffic data

D. CPU usage statistics

Answer: B. A script of actions choreographed in the GUI

VTUSYNC.IN